

The Agentic-Researcher on ASU Sol HPC Cluster

Using Codex CLI with ChatGPT Plus

Overview

This guide explains how to run The-Agentic-Researcher on ASU Sol using:

- Apptainer
- Codex CLI
- ChatGPT Plus authentication

This setup does **not** require:

- Ollama
- OpenCode
- Local LLM hosting
- API keys
- GPU resources solely for accessing the model

Codex runs through your ChatGPT Plus account while Agentic Researcher provides sandboxing, planning, execution, and project management.

Prerequisites

- Active ASU Sol account
- SSH access to Sol
- ChatGPT Plus account
- The-Agentic-Researcher repository installed

```
# Part 0: Get The-Agentic-Researcher from GitHub
```

If you have not already cloned the repository, do this once on Sol:

```
```bash
cd ~
git clone https://github.com/ZIB-IOL/The-Agentic-Researcher.git
cd The-Agentic-Researcher
```

If the repository is already present, update it with:

```
cd ~/The-Agentic-Researcher
git pull
```

You should now have the project directory at:

```
~/The-Agentic-Researcher
```

Then the rest of the flow stays the same:

```
```bash
curl -fsSL https://chatgpt.com/codex/install.sh | sh
codex login --device-auth
agentic-researcher --setup
agentic-researcher --apptainer
```

Part 1: Install Codex CLI

SSH into Sol:

```
ssh hwang49@sol-login05.asu.edu
```

Install Codex:

```
curl -fsSL https://chatgpt.com/codex/install.sh | sh
```

Verify installation:

```
which codex
```

```
codex --help
```

Expected output:

```
/home/hwang49/.local/bin/codex
```

Part 2: Authenticate Codex

Because Sol is accessed through SSH, use device authentication.

Run:

```
codex login --device-auth
```

You will see:

```
https://auth.openai.com/codex/device
```

ABCD-EFGH

Open the URL on your local computer.

Sign in with your ChatGPT Plus account.

Enter the code.

Wait for authentication to complete.

Important

Authentication is normally required only once.

Credentials are stored in your home directory and remain available across SSH sessions and login nodes.

You normally do not need to log in again unless:

- You run `codex logout`
- Your credentials expire
- Your home directory is reset
- The Codex configuration is removed

Part 3: Configure Agentic Researcher

Run:

```
agentic-researcher --setup
```

Select:

Container Runtime:

3) `apptainer`

CLI Tool:

4) `codex`

For most users:

Default model:

<leave blank>

Proxy:

<leave blank>

State/cache directory:

<accept default>

Extra Sandbox Directories

When prompted:

Extra Sandbox Directories

add any locations containing datasets or projects.

Example:

/home/hwang49/research

/home/hwang49/scratch

/packages

This allows Codex to access those locations without repeatedly requesting permission.

Configuration is saved to:

~/.config/agentik-researcher/config.sh

Verify:

grep AR_CLI_TOOL ~/.config/agentik-researcher/config.sh

Expected:

AR_CLI_TOOL=codex

Part 4: Build Apptainer Image

Only required once.

From the repository:

```
cd ~/The-Agentic-Researcher
```

```
agentic-researcher --apptainer --build
```

This step may take a long time.

After the image is built, you normally never need to rebuild it.

Changing from Claude, OpenCode, or Gemini to Codex does not require rebuilding.

Part 5: Launch Agentic Researcher

Always start inside a project directory.

Example:

```
cd ~/research/quantum_feature_selection
```

```
agentic-researcher --apptainer
```

Avoid launching from:

```
cd ~
```

```
agentic-researcher --apptainer
```

because this makes your entire home directory the workspace and can trigger unnecessary permission prompts.

First Startup

You should see:

```
Agentic Researcher - Codex CLI
```

```
Starting Codex CLI...
```

If prompted:

```
Yes, I trust this folder
```

select:

Yes

Initial Research Setup

Inside Agentic Researcher:

/setup_research_plan

The framework will create:

report.tex

TODO.md

AGENTS.md

These files track:

- Research progress
- Planned tasks
- Project instructions
- Experiment history

Daily Usage

After setup, a typical session is:

```
ssh hwang49@sol-login05.asu.edu
```

```
cd ~/research/project_name
```

```
agentic-researcher --apptainer
```

No additional login is normally required.

Using ASU GPUs

When a GPU Is NOT Needed

You do NOT need a GPU simply to use Codex.

The model runs on OpenAI infrastructure through your ChatGPT Plus account.

The following works entirely on a login node:

```
ssh sol-login05
```

```
cd ~/research/project
```

```
agentic-researcher --apptainer
```

When a GPU IS Needed

Request a GPU when your project performs computation such as:

- PyTorch training
- TensorFlow training
- Fine-tuning LLMs
- Running Hugging Face models
- CUDA applications
- Quantum machine learning experiments
- Large-scale numerical simulations

Request a GPU node:

```
interactive -t 8:00:00 -G 1 -p htc
```

or

```
srun --pty -t 8:00:00 -G 1 -p htc bash
```

Then:

```
cd ~/research/project
```

agentic-researcher --apptainer

Codex can then generate and execute code using the allocated GPU resources.

Understanding Sandbox Warnings

You may see messages such as:

The sandbox failed before running a basic workspace inspection command

or

Would you like to run this command?

This is normal.

Agentic Researcher runs in:

Protected Mode

and asks permission before executing commands.

Typical startup commands are harmless:

pwd

rg --files

git status

These commands only inspect the workspace.

Choose:

Yes, proceed

for normal workspace inspection.

Common Commands

Initialize research:

/setup_research_plan

Show help:

/help

Compact conversation history:

/compact

Exit:

/exit

Troubleshooting

Codex Not Found

Check:

which codex

If missing:

```
curl -fsSL https://chatgpt.com/codex/install.sh | sh
```

Authentication Requested Again

Run:

```
codex login --device-auth
```

and repeat the browser authentication process.

Agentic Researcher Starts Claude Instead of Codex

Run:

```
agentic-researcher --setup
```

Choose:

4) codex

Verify:

```
grep AR_CLI_TOOL ~/.config/agentic-researcher/config.sh
```

Expected:

AR_CLI_TOOL=codex

Recommended Workflow for Research

Login:

```
ssh hwang49@sol-login05.asu.edu
```

If GPU computation is needed:

```
interactive -t 8:00:00 -G 1 -p htc
```

Move to the project:

```
cd ~/research/my_project
```

Launch:

```
agentic-researcher --apptainer
```

Initialize:

```
/setup_research_plan
```

Then let Codex manage:

- Literature review
- Coding
- Experiment planning
- Data analysis
- Report generation
- Project tracking
- Reproducible workflows

using ChatGPT Plus as the reasoning engine and ASU Sol as the computational platform.